

Assignment 2

(Due on Apr. 12, 2026)

Question 1 光电效应

A photoelectric experiment indicates that green light of wavelength 5220 Å is the longest wavelength radiation that can cause the photoemission of electron from a clean sodium surface. (a) What is the workfunction of sodium, in eV?

(b) If a UV radiation of wavelength 2500 Å incident upon the sodium surface, what will be the maximum kinetic energy of the photoemitted electrons, in eV?

(a) 材料截止波长 $\lambda_0 = 5220 \text{ Å}$

$$\Phi = \frac{hc}{\lambda_0} \quad hc = 12400 \text{ eV} \cdot \text{Å} \quad \Rightarrow \quad \Phi = \frac{12400 \text{ eV} \cdot \text{Å}}{5220 \text{ Å}} \approx 2.3755 \text{ eV}$$

(b) $E = \frac{hc}{\lambda} = \frac{12400 \text{ eV} \cdot \text{Å}}{2500 \text{ Å}} = 4.96 \text{ eV}$

$$K_{\max} = E - \Phi$$

$$K_{\max} = 4.96 \text{ eV} - 2.3755 \text{ eV} = 2.5845 \text{ eV}$$

Question 2: 光电效应

推断 Deduce the work function for sodium from Fig. Q2. The intersection of the straight line with the horizontal axis is the cutoff frequency, $\nu_0 = 5.6 \times 10^{14} / \text{sec}$. For most conducting metals the value of the work function is of the order of a few electron volts. Calculate the photon energy of a typical microwave wavelength of $\lambda = 10 \text{ cm}$. Can this energy sufficient to eject photoelectrons from metal surfaces?

功函数的值大约在几个电子伏特的数量级

赶出, 喷射, 排出

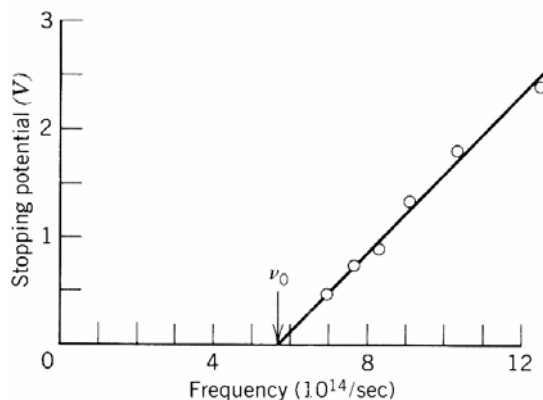


Fig. Q2

截止频率 $\nu_0 = 5.6 \times 10^{14} \text{ Hz}$

$$\Phi = h\nu_0 = (4.136 \times 10^{-15} \text{ eV} \cdot \text{s}) \times (5.6 \times 10^{14} \text{ s}^{-1}) \approx 2.316 \text{ eV}$$

$$E = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34} \text{ J} \cdot \text{s} \times 3 \times 10^8 \text{ m/s}}{0.1 \text{ m} \times 1.602 \times 10^{-19} \text{ J/eV}} \approx 1.24 \times 10^{-5} \text{ eV}$$

显然 $E \ll \Phi$

无法激发光电子

$$h \approx 6.626 \times 10^{-34} \text{ J} \cdot \text{s}$$

$$\approx 4.136 \times 10^{-15} \text{ eV} \cdot \text{s}$$

$$hc \approx 1.24 \times 10^{-6} \text{ eV}$$

$$1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$$

$$1 \text{ Å} = 10^{-10} \text{ m}$$

Question 3: 波粒二象性

Calculate the wavelength associated with 1 eV (a) photon (b) electron.

(a) 光子 $E = \frac{hc}{\lambda} \Rightarrow \lambda = \frac{hc}{E} = \frac{1.24 \times 10^{-6} \text{ eV}}{1 \text{ eV}} = 1240 \text{ nm}$

(b) 电子 $\lambda = \frac{h}{p}$
 $k = \frac{p}{\hbar} \Rightarrow p = \hbar k$
 $\Rightarrow \lambda = \frac{h}{\hbar k} = \frac{6.626 \times 10^{-34}}{\sqrt{2 \times 9.11 \times 10^{-31} \times 1.602 \times 10^{-19}}} \approx 1.24 \text{ nm}$

低能电子 (1 eV 远小于电子静能 511 keV)

可用经典力学公式

速解: 低能电子, $\lambda(\text{nm}) \approx \sqrt{\frac{1.5}{V}}$

当能量为 1 eV 时, $V = 1 \text{ v}$

∴ 在半导体分析中

高能电子束比可见光观测有更高的分辨率

Question 4 势垒问题

波数

势能

A free particle of mass m with wave number k_1 is traveling to the right. At $x = 0$, the potential jumps from zero to U_0 and remains at this value for positive x .

(a) If the total energy is $E = \frac{\hbar^2 k_1^2}{2m} = 2U_0$, what is the wave number k_2 in the region $x > 0$? Express your answer (i) in terms of U_0 and (ii) in terms of k_1 .

(b) Calculate the reflection coefficient R at the potential step.

(c) What is transmission coefficient T ?

(a) 求 $x > 0$ 区域波数 k_2

(i) 用 U_0 表示

$$E = 2U_0 \quad k_2 = \sqrt{\frac{2m(E - U_0)}{\hbar^2}} = \sqrt{\frac{2mU_0}{\hbar^2}}$$

(ii) 用 k_1 表示

$$\frac{\hbar^2 k_1^2}{2m} = 2U_0 \Rightarrow U_0 = \frac{\hbar^2 k_1^2}{4m} \quad k_2 = \sqrt{\frac{2m}{\hbar^2} \cdot \frac{\hbar^2 k_1^2}{4m}} = \frac{k_1}{\sqrt{2}}$$

(b) 算反射系数 R

$$\text{对 } E > U_0 \text{ 的势阶 } R = \left(\frac{k_1 - k_2}{k_1 + k_2} \right)^2 = \left(\frac{1 - \frac{1}{\sqrt{2}}}{1 + \frac{1}{\sqrt{2}}} \right)^2 = (3 - 2\sqrt{2})^2 = 17 - 12\sqrt{2} \approx 0.032$$

即 3.2% 的粒子被反射

(c) 算透射系数 T

$$T = 1 - R = 12\sqrt{2} - 16 \approx 0.968$$

即 96.8% 的粒子透射入区域 2

Question 5 量子隧穿

Consider the phenomenon of tunneling through a potential energy barrier of height V_0 and width a , as shown in the figure below.

- Find the reflection coefficient R ?
- What happens to R as a or V_0 or both become very large?
- For a wide barrier ($ka \gg 1$), show that T_0 can at most be 4 and that $T_0 = 4$ when $E = (1/2)V_0$.

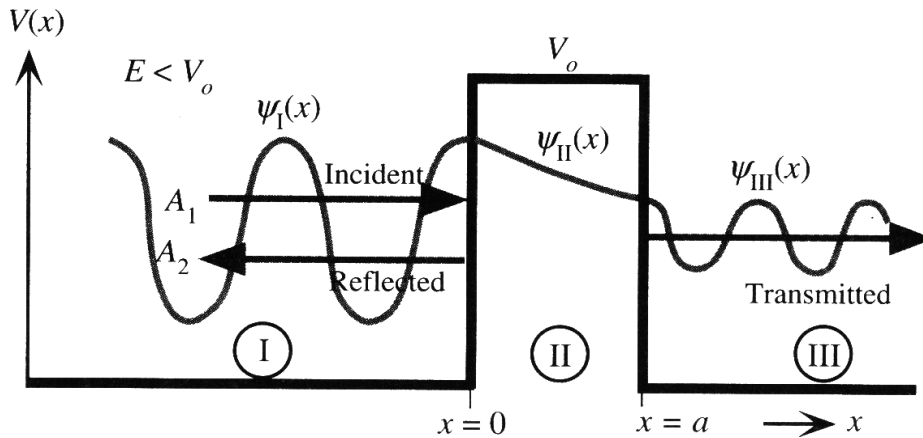


Fig. Q5

(a) 求反射系数 R

通过边界条件, 推导出 T 的标准表达式

$$T = \frac{1}{1 + \frac{V_0^2}{4E(V_0 - E)} \sinh^2(ka)}$$

$$\text{代入 } R = 1 - T \Rightarrow R = \frac{V_0^2 \sinh^2(ka)}{4E(V_0 - E) + V_0^2 \sinh^2(ka)}$$

$$\text{其中 } k = \sqrt{\frac{2m(V_0 - E)}{\hbar^2}}$$

(b) 当 a 和 V_0 非常大时, R 如何变

当 $a \rightarrow \infty$ 或 $V_0 \rightarrow \infty$ 时, $ka \rightarrow \infty$

当 $ka \gg 1$ 时, $\sinh^2(ka) \approx \frac{1}{4}e^{2ka} \rightarrow \infty$

当 $V_0^2 \sinh^2(ka) \gg 4E(V_0 - E)$ 时, $R \rightarrow 1$

即势垒变得无限宽或无限高时, 粒子无法隧穿, 全反射

(c) 对宽势垒 ($ka \gg 1$)

当 $ka \gg 1$ 时, $\sinh^2(ka) \approx \frac{1}{4}e^{2ka}$

$$\Rightarrow T \approx \frac{1}{\frac{V_0^2}{4E(V_0 - E)} \frac{1}{4}e^{2ka}} = \left[\frac{16E(V_0 - E)}{V_0^2} \right] e^{-2ka}$$

$$\text{令 } x = \frac{E}{V_0} \quad (0 < x < 1) \quad T_0(x) = \frac{16E(V_0 - E)}{V_0^2} = 16x(1-x)$$

$$\text{当 } x = \frac{1}{2} \text{ 时 } T_{0\max} = 16 \cdot \frac{1}{2} \cdot (1 - \frac{1}{2}) = 4$$

1. 薛定谔方程:

$$(\text{定态}) -\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + U(x)\psi = E\psi$$

2. 波数 k : 波函数的空间振荡频率

$$k = \sqrt{\frac{2m(E - U)}{\hbar^2}}$$

3. 波函数连续性条件:

在势能突变的边界 ($x=0$), 波函数 $\psi(x)$ 及 $\psi'(x)$ 必须连续

4. 区域波函数:

$$\begin{cases} \text{区域 I } (x < 0, V=0) & \psi_I(x) = A_1 e^{ikx} + A_2 e^{-ikx} & k = \sqrt{\frac{2mE}{\hbar^2}} \\ \text{区域 II } (0 \leq x \leq a, V=V_0) & \psi_{II}(x) = B_1 e^{kx} + B_2 e^{-kx} & k = \sqrt{\frac{2m(V_0 - E)}{\hbar^2}} \\ \text{区域 III } (x > a, V=0) & \psi_{III}(x) = C e^{ikx} \end{cases}$$

5. 反射系数 R 与透射系数 T :

$$\text{概率流密度之比 } \begin{cases} R = \frac{|A_2|^2}{|A_1|^2} \\ T = \frac{|C|^2}{|A_1|^2} \end{cases}$$

$$\text{对势阶问题 } R = \left(\frac{k_1 - k_2}{k_1 + k_2} \right)^2$$

$$\text{守恒律 } R + T = 1$$